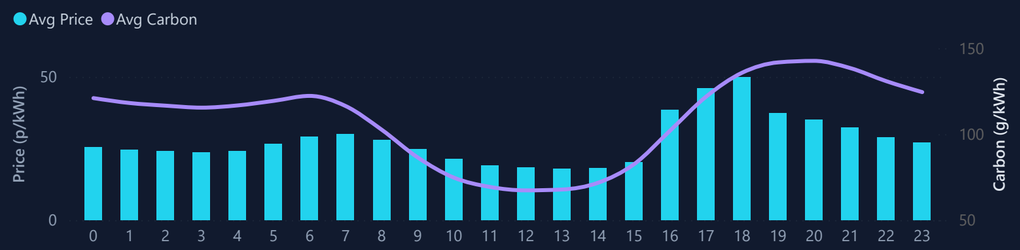
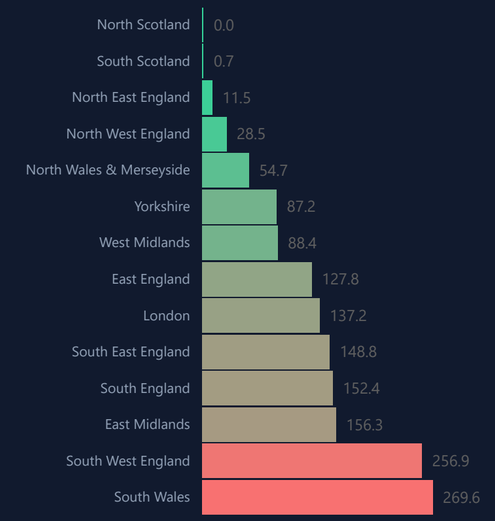
UK Grid Live: When is electricity cleanest and cheapest?

Avg Carbon	Avg Price	Wind Share %	Peak Premium %
108.6	27.9	38%	76%

Midday is cleanest and cheapest; 4–7pm costs the most



Carbon by region (g/kWh)



Region

All

Date range

25/07/2026 — 23/10/2026

Negative Price Label
2,590

More wind = cleaner and cheaper power

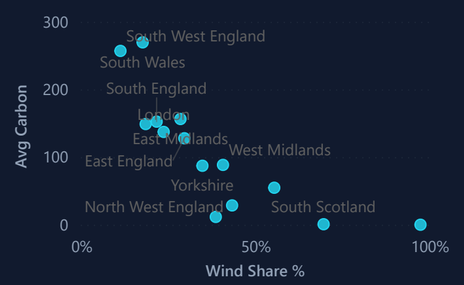
Carbon when windy (40%+ wi...	Carbon when calm (<15% wind)	Price when windy (p)	Price when calm (p)
35.5	205.7	25.7	30.8

Cheapest times: hour × day (p/kWh)

day_name	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	23
Monday	27.1	26.5	26.3	26.1	27.2	31.3	34.6	36.0	34.4	30.0	26.7	24.2	23.1	22.5	22.6	25.4	43.0	49.1	52.3	38.5	35.7	33.2	29.7	28.5
Tuesday	27.2	26.1	25.7	25.3	25.9	28.3	32.4	33.6	30.8	27.6	24.2	21.1	20.4	19.8	20.0	22.8	40.5	49.1	53.7	41.1	37.0	33.6	29.4	26.7
Wednesday	25.3	24.9	24.7	24.4	25.1	29.0	32.7	34.1	30.6	27.9	25.0	22.9	21.8	21.5	21.9	23.2	40.6	46.2	51.0	38.5	35.9	32.6	28.7	27.0
Thursday	25.4	24.0	23.2	22.9	23.6	26.4	30.1	31.6	30.2	27.2	22.5	19.8	18.9	18.7	19.3	21.8	39.8	46.4	51.2	39.5	36.9	33.5	29.9	28.0
Friday	26.0	24.7	24.1	23.9	24.4	27.4	31.0	32.6	29.8	26.5	23.1	20.9	20.0	19.4	19.3	19.9	37.2	43.7	46.8	34.1	32.2	29.4	25.9	23.5
Saturday	21.8	20.5	19.9	19.2	19.5	19.8	20.3	19.8	18.8	15.5	11.6	9.9	9.6	9.2	9.4	11.5	31.9	41.9	45.2	33.1	31.9	30.5	28.0	26.5
Sunday	25.5	24.8	23.7	22.8	22.7	22.7	22.6	21.8	20.0	18.0	16.3	14.6	14.1	13.8	14.3	15.9	36.5	45.1	48.2	36.1	35.2	33.2	30.2	28.2

Windier regions are cleaner

Wind Share % and Avg Carbon by region_na...



Carbon falls when the wind blows (daily)

Avg Carbon and Wind Share % by local_date

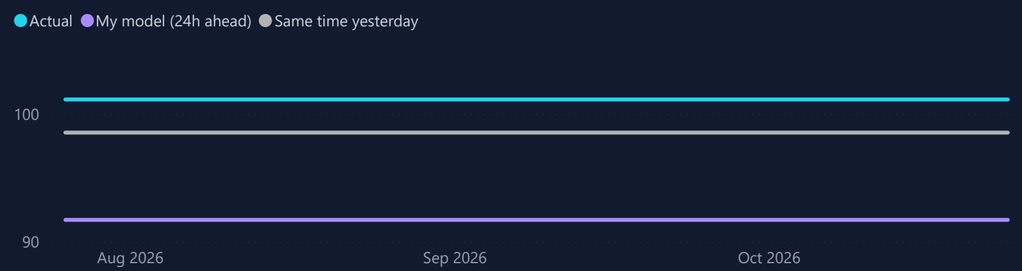


Can we predict tomorrow’s carbon?

My model error (g)	Same-time-yesterday error (g)	National Grid error (g)	Improvement vs naive
22.4	38.8	10.1	42%

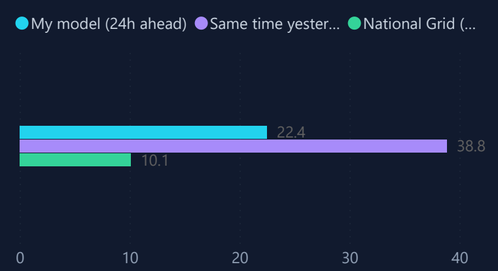
24-hour-ahead forecast vs actual (last 14 days)

Actual, My model (24h ahead) and Same time yesterday by utc_datetime



Average error (gCO2/kWh, lower is better)

My model (24h ahead), Same time yesterday and ...



How the model works: gradient boosting (scikit-learn) trained on 75 days, tested on the last 14. Inputs are only things known 24 hours ahead: offshore wind forecasts (Dogger Bank, Irish Sea, Aberdeen), solar and temperature forecasts, time of day, day of week, and carbon from 1, 2 and 7 days earlier.

Top drivers: Dogger Bank wind and average wind speed, then time of day.

Honest comparison: National Grid’s archived forecast is updated right up to delivery, so it is a reference rather than a like-for-like competitor. My model forecasts a full day ahead and cuts error by 42% versus a same-time-yesterday baseline.